

Figure 1: OpenDaylight-OpenStack

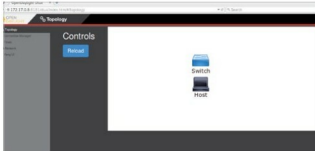


Figure 2: Dlxr login screenshot

```
$ tail -f data/log/karaf.log
$ tail -f logs/web_access_log_2015-12.txt
```

Now, you have a working OpenDaylight-Lithium set-up; so, let's get into the integration part.

3. Erase all VMs, networks, routers and ports in the controller node

Since you already have a working OpenStack set-up, you may test for VM provisioning, but before integrating OpenStack with OpenDaylight, you must clean up all the unwanted data from the OpenStack database. When using OpenDaylight as the Neutron back-end, ODL expects to be the only source for Open vSwitch configuration. Because of this, it is necessary to remove existing OpenStack and Open vSwitch (OVS) configurations to give OpenDaylight a clear field.

The following steps will guide you through the cleaning process:

- Delete the instances


```
$ nova list
$ nova delete <instance names>
```
- Remove links from subnets to routers


```
$ neutron subnet-list
$ neutron router-list
$ neutron router-port-list <router name>
$ neutron router-interface-delete <router name> <subnet ID or name>
```
- Delete subnets, nets, routers

```
$ neutron subnet-delete <subnet name>
$ neutron net-list
$ neutron net-delete <net name>
$ neutron router-delete <router name>
```

- Check that all ports have been cleared - at this point, this should be an empty list


```
$ neutron port-list
```

While Neutron is managing the OVS instances on compute and control nodes, OpenDaylight and Neutron can be in conflict. To prevent issues, turn off the Neutron server on the network controller and Neutron's Open vSwitch agents on all hosts.

```
$ service neutron-server stop
```

4. Configuring Open vSwitches in network and compute nodes

The Neutron plugin in every node must be removed because only OpenDaylight will be controlling the Open vSwitches. So on each host, we will clear the pre-existing Open vSwitch configuration and set OpenDaylight to manage the Open vSwitch.

```
$ apt-get purge neutron-plugin-openvswitch-agent
$ service openvswitch-switch stop
$ rm -rf /var/log/openvswitch/*
$ rm -rf /etc/openvswitch/conf.db #Clear openvswitch
database and start it again.
$ service openvswitch-switch start
$ ovs-vsctl show #This command must return the empty set
except OpenVswitch ID and
it's Version.
```

5. Connecting Open vSwitch with OpenDaylight

Use the following commands to make OpenDaylight manage Open vSwitch:

```
$ ovs-vsctl set Open_vSwitch <OPENVSWITCH ID> other_
config={'local_ip'='<TUNNEL INTERFACE IP>'}
$ ovs-vsctl set-manager tcp:<OPENDAYLIGHT MANAGEMENT IP>:6640
```

You can get the Open vSwitch ID from the command `ovs-vsctl show` and the tunnel IP from the OpenStack `ml2_conf.ini` file.

Execute the above command in all the nodes (network and compute nodes) that have Open vSwitch installed.

Create the bridge `br-ex`, which is needed for the external network for OpenStack in the network (Neutron) node.

```
$ ovs-vsctl add-br br-ex
$ ovs-vsctl add-port br-ex <INTERFACE NAME OF EXTERNAL NETWORK>
#commonly eth0 or p2p1
$ ovs-vsctl show
```

The above command will show that you are connected to the